Task Optimization Drives Statistical Decorrelation: An Empirical Study of Integration Dynamics in Feed-forward and Recurrent Neural Networks

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Abstract

Integrated Information Theory (IIT) posits that consciousness is related to a system's capacity for information integration (Φ). In contrast, modern deep learning systems are optimized for task performance, often through mechanisms that promote feature disentanglement and redundancy reduction. In this study, we investigate the relationship between task optimization and internal statistical integration by analyzing the training dynamics of Feed-forward (MLP) and Recurrent Neural Networks (RNN) on non-linearly separable tasks (e.g., XOR).

Using Gaussian Total Correlation (TC) as a tractable statistical proxy for multivariate dependency (not a direct measure of Φ), we observe a consistent inverse relationship: as model accuracy improves, internal information redundancy significantly decreases. Furthermore, we find that recurrent architectures maintain higher baseline levels of statistical integration compared to feed-forward models, likely due to temporal feedback mechanisms, yet still exhibit a similar decay trend during optimization.

These results suggest a potential trade-off between task performance and statistical integration in standard neural architectures. While not directly measuring integrated information in the IIT sense, our findings highlight a systematic tendency toward decorrelation under gradient-based optimization, which may have implications for integration-based theories of consciousness. This work underscores the importance of temporal feedback and architectural design in sustaining integrated representations during learning.

Keywords*: Statistical Integration, Total Correlation, Neural Networks, Training Dynamics, Representation Learning, Recurrent Networks*

1. INTRODUCTION

The quest to understand the relationship between intelligence, defined as the capacity to optimize for specific goals and consciousness often associated with subjective experience remains one of the central challenges in cognitive science and artificial intelligence. Integrated Information Theory (IIT) provides a formal framework for consciousness, proposing that a system s level of consciousness is related to its capacity to integrate information in a manner that is irreducible to its parts, quantified by Φ [1], [2].

Concurrently, modern Deep Learning (DL) has achieved remarkable success by optimizing neural networks for task performance. However, theoretical perspectives from Information Bottleneck and representation learning suggest that effective generalization involves compressing input data and reducing redundant correlations [3], [4]. This introduces a potential tension: does the optimization process underlying "intelligence" (task efficiency) inherently support or suppress the high levels of integration associated with consciousness in IIT?

In this work, we empirically investigate this question by analyzing the training dynamics of two neural architectures Feed-forward Neural Networks (MLP) and Recurrent Neural Networks (RNN) on non-linearly separable tasks. Rather than attempting to directly compute the computationally intractable causal quantity Φ, we employ Gaussian Total Correlation (TC) as a tractable statistical proxy for multivariate dependency (not a direct measure of Φ) among hidden representations.

We hypothesize that standard gradient-based optimization drives neural networks toward progressively decorrelated internal representations, suggesting a potential trade-off between statistical integration and task performance [5].

2. RELATED WORK

To contextualize our findings on the relationship between task optimization and statistical integration, we draw upon three complementary research areas: theories of consciousness, information-theoretic analyses of deep learning, and studies of representation decorrelation.

1. Integrated Information Theory (IIT)

Integrated Information Theory, proposed by Tononi, formalizes consciousness as a property of systems with high levels of irreducible information integration, quantified by Φ [1], [2]. While IIT provides a rigorous theoretical framework, computing Φ for high-dimensional and continuous systems such as neural networks is computationally intractable [6], [7]. Prior work (e.g., Mediano et al., Barrett) has explored approximations, yet empirical studies of integration dynamics during neural network training remain limited. In this work, we adopt Gaussian Total Correlation (TC) as a tractable statistical proxy for multivariate dependency, allowing us to analyze integration-related behavior during learning.

2. Information Bottleneck and Representation Compression

The Information Bottleneck framework (Tishby et al.) provides a theoretical perspective on deep learning as a process of information compression [3], [4]. Neural networks are hypothesized to minimize irrelevant input information while preserving predictive features. Empirical studies (e.g., Shwartz-Ziv & Tishby, 2017) suggest that training consists of an initial fitting phase followed by a compression phase [5].
Our findings are consistent with this perspective: we observe a systematic reduction in redundancy (as measured by TC) during training. However, we extend this interpretation by analyzing how this reduction relates to statistical integration, providing a complementary view of representation dynamics.

3. Decorrelation and Efficient Representations

A substantial body of work in deep learning emphasizes the importance of decorrelation and redundancy reduction. Techniques such as Dropout, Batch Normalization, and explicit regularization methods (e.g., DeCov loss) aim to reduce internal dependencies and improve generalization [8].
This literature establishes decorrelation as a key property of efficient representations. Our contribution is to quantify this phenomenon over training and compare its behavior across architectures, showing that while decorrelation improves performance, it is associated with a reduction in statistical integration.

To our knowledge, prior work has not systematically quantified the relationship between task performance and statistical integration across neural architectures during training. This study aims to fill this gap by providing an empirical analysis of integration dynamics in both feed-forward and recurrent networks.

3. PROPOSED METHODOLOGY

3.1. Experimental Objective

The objective of this study is to investigate how task optimization affects internal statistical integration in neural networks. Specifically, we analyze whether increasing task performance through gradient-based learning is associated with a reduction in multivariate statistical dependency among hidden units.

We compare two architectures: a feed-forward Multi-Layer Perceptron (MLP) and a Recurrent Neural Network (RNN). The MLP represents a purely spatial feed-forward computation, while the RNN introduces temporal feedback through recurrent hidden-state updates. This comparison allows us to examine whether recurrent feedback helps preserve statistical integration during learning.

3.2. Task and Dataset: Continuous XOR

We use the XOR task as a minimal non-linearly separable benchmark. XOR is suitable for this study because solving it requires the model to learn interactions between input dimensions rather than relying on independent linear separability. Instead of using the discrete binary XOR truth table, we generate a continuous noisy XOR dataset. Input samples are two-dimensional continuous vectors drawn from four clusters corresponding to the XOR quadrants.

Gaussian noise is added to each input coordinate to create a stochastic continuous manifold. This continuous formulation serves two purposes. First, it makes the task more appropriate for neural network training. Second, it enables stable estimation of covariance-based statistical quantities, which are required for Gaussian Total Correlation.
The dataset consists of $N=1024$ samples. The data are split into training and evaluation sets, and model performance is measured using classification accuracy.

3.3. Neural Architectures

3.3.1 Feed-forward Network

The feed-forward model is a compact Multi-Layer Perceptron composed of fully connected layers with Tanhactivations. Information flows unidirectionally from input to hidden layers and finally to the output layer:

$$h = \tanh(Wx + b) \quad (1)$$

This architecture has no recurrent connections or temporal memory. Therefore, any measured statistical integration reflects spatial dependency among hidden units within a layer.

3.3.2 Recurrent Network

The recurrent model is a vanilla RNN based on recurrent hidden-state updates. At each time step, the hidden state depends on both the current input and the previous hidden state:

$$h_t = \tanh(W_{in}x_t + W_{rec}h_t - 1 + b) \quad (2)$$

The model is unrolled for a fixed number of time steps. This recurrent structure introduces temporal feedback, allowing hidden states to depend on previous internal activity. Recurrent connectivity may support richer internal dependencies and is therefore relevant to studying integration-related dynamics.

Importantly, we do not claim that recurrence directly implies IIT-style integrated information. Rather, recurrence is used here as an architectural mechanism that can increase temporal coupling among internal representations.

3.4. Activation Function

Both architectures use the Tanh activation function. This choice bounds neural activations within $[-1,1]$, reducing the risk of extreme variance and improving numerical stability during covariance estimation.
Tanh is also more suitable than ReLU for this experiment because ReLU produces semi-bounded sparse activation distributions, which can distort Gaussian covariance-based entropy approximations.

3.5. Measuring Statistical Integration Using Gaussian Total Correlation

Directly computing IIT s causal integrated information Φ for continuous high-dimensional neural networks is computationally intractable. Therefore, we use Gaussian Total Correlation (TC) as a tractable statistical proxy for multivariate dependency among hidden units.

Total Correlation measures the difference between the sum of marginal entropies and the joint entropy of a multivariate system [9]:

$$TC(X) = \sum_{i=1}^{N} H(X_i) - H(X_1, \ldots, X_N) \quad (3)$$

where X_i represents the activity of the $i - th$ hidden unit. Under a multivariate Gaussian approximation, TC can be computed using the covariance matrix Σ:

$$TC(X) = \frac{1}{2} \ln \left(\frac{\prod_{i=1}^{N} \Sigma_{ii}}{\det(\Sigma_i)} \right) \quad (4)$$

where Σ_{ii} is the variance of the $i-th$ unit and $\det(\Sigma)$ is the determinant of the full covariance matrix.

High TC indicates stronger statistical coupling and redundancy among hidden units. Low TC indicates weaker dependency, suggesting decorrelation and specialization of internal representations. We emphasize that TC is not equivalent to IIT s Φ.

TC measures statistical dependency, while Φ measures causal irreducibility. In this study, TC is used only as an operational approximation for tracking integration-related statistical structure during training.

3.6. Numerical Stability

To ensure stable computation of the log-determinant, especially when covariance matrices are nearly singular, we add a small diagonal regularization term:

$$\Sigma_{stable} = \Sigma + \epsilon I \quad (5)$$

Where $\epsilon = 10^{-6}$

The stable TC estimate is computed as:

$$TC_{stable} = \frac{1}{2}\left[\sum_{i=1}^{N} \left(\ln\left(\sum_{ii}\right) - \ln \det\left(\sum + \epsilon I\right)\right)\right] \quad (6)$$

This prevents numerical instability caused by collinear or highly correlated hidden activations.

3.7. Training Protocol

Both models are trained using the Adam optimizer with a learning rate of 0.01. The objective function is Cross-Entropy Loss, and training is performed for 40 epochs. At the end of each epoch, the model is switched to evaluation mode.

The evaluation dataset is passed through the network, and hidden activations are collected. These activations are then used to estimate the covariance matrix and compute epoch-wise Gaussian TC.

This procedure produces a temporal trajectory of both task performance and statistical integration:

$$Accuracy_t, TC_t$$

where t denotes the training epoch.

Experiments are repeated across multiple random seeds to assess robustness. Reported results should include mean and Standard deviation across seeds whenever possible.

3.8. Analysis Strategy

We analyze three main relationships:

- The relationship between accuracy and TC during training.
- The rate at which TC decreases as task performance improves.
- The difference between feed-forward and recurrent architectures in maintaining statistical integration.

The central hypothesis is that task optimization drives internal representations toward decorrelation. Therefore, we expect accuracy to increase while TC decreases. We further hypothesize that recurrent architectures will maintain higher baseline TC than feed-forward architectures because of temporal feedback, while still showing a decay trend under optimization.

4. RESULTS AND EVALUATION

We present an empirical analysis of training dynamics in both feed-forward (MLP) and recurrent (RNN) architectures. Our focus is on the joint evolution of task performance (accuracy) and statistical integration (Gaussian Total Correlation, TC) in order to evaluate the hypothesis that gradient-based optimization is associated with progressive decorrelation of internal representations.

4.1. Training Dynamics in Feed-forward Networks

We observe a consistent inverse relationship between task performance and statistical integration in the MLP model. As shown in Figure 1, training proceeds in two distinct regimes:

Early Phase (Epochs 0 - 5).

At initialization, the model exhibits relatively high statistical integration (TC≈35.3 nats) and low classification accuracy (~55%).This indicates strong statistical dependency among hidden units.

Optimization Phase (Epochs 6 - 40).

As training progresses, accuracy increases to approximately 88%, While TC decreases substantially to approximately 16.0 nats, corresponding to a reduction of over 50%. This result suggests that performance improvements are associated with a Progressive reduction in statistical dependency among hidden units.

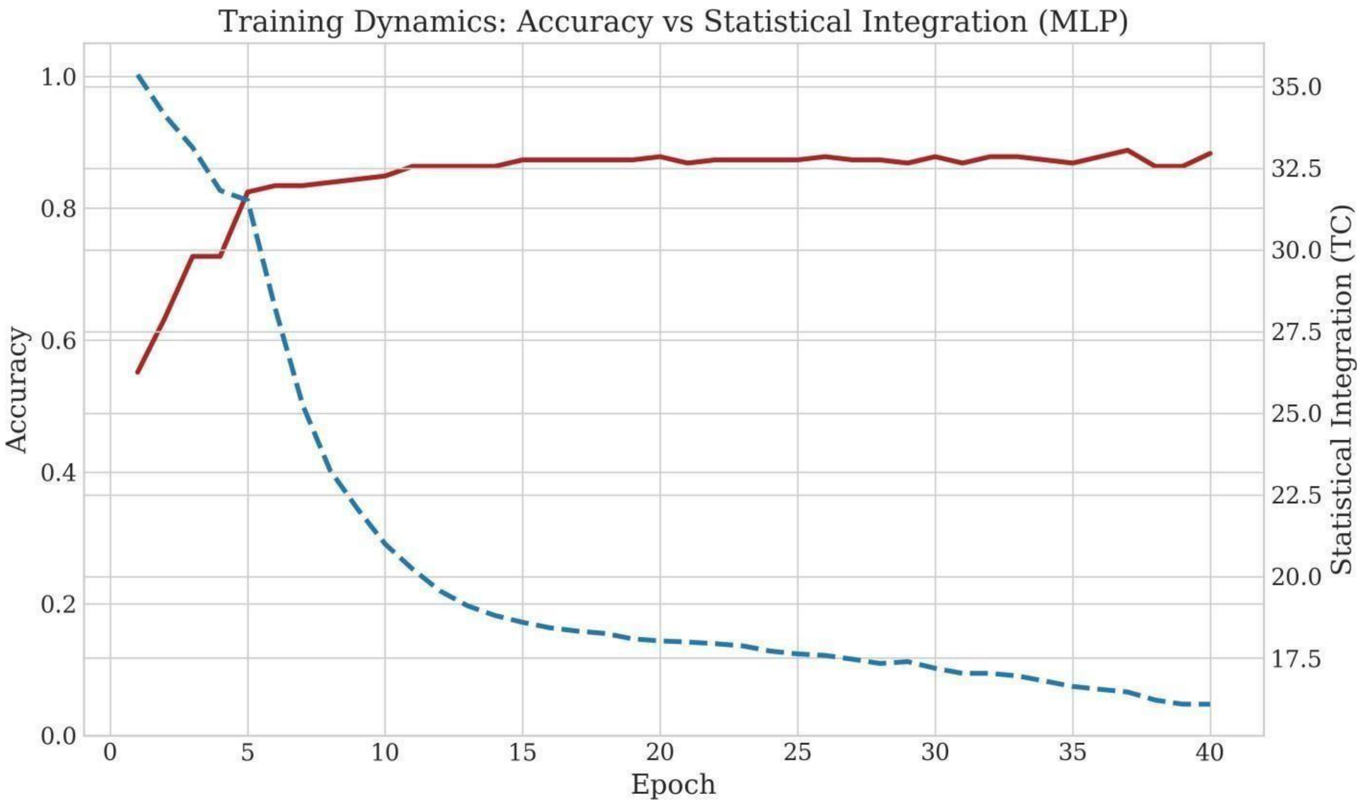

Figure 1. Training Dynamics: Accuracy vs Statistical Integration (MLP)

As shown in Figure 1, training exhibits a clear inverse relationship between accuracy and Total Correlation (TC). During early training, the model maintains high TC with relatively low accuracy. As optimization progresses, accuracy increases while TC decreases substantially, indicating progressive decorrelation.

4.2. Layer-wise Analysis (MLP)

To better understand how decorrelation develops within the network, we analyze layer-wise Total Correlation across training.

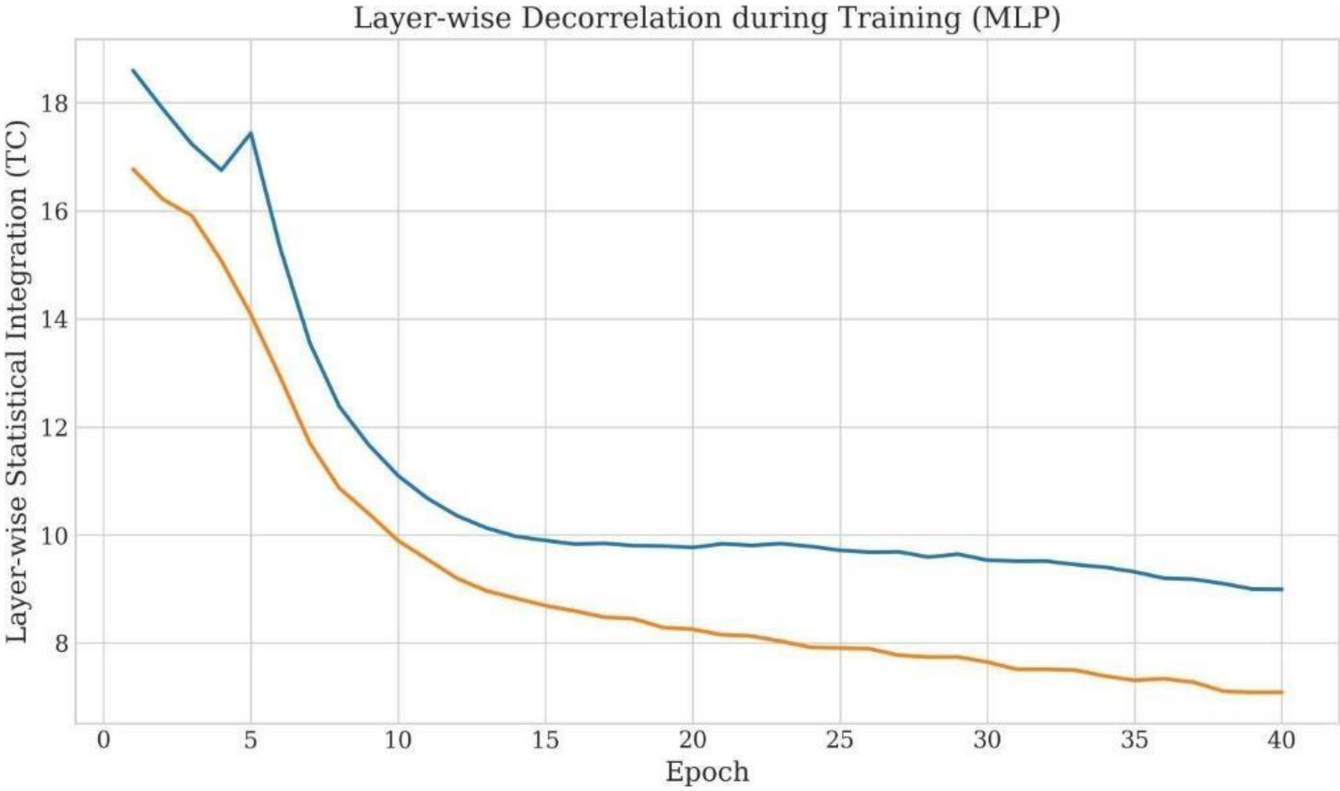

Figure 2. Layer-wise Decorrelation During Training (MLP)

Figure 2 shows that decorrelation occurs across all layers, with deeper layers exhibiting stronger reductions in statistical dependency. This suggests increasing specialization of representations as depth increases.

4.3 Architecture Comparison (MLP vs RNN)

We now compare the statistical integration dynamics between feed-forward and recurrent architectures. Figure 3 illustrates the evolution of Total Correlation (TC) across training epochs for both MLP and RNN models. Both architectures exhibit a progressive reduction in TC during optimization, indicating increasing decorrelation of internal representations. However, the recurrent architecture consistently maintains higher TC values throughout training compared to the feed-forward model. This suggests that temporal feedback mechanisms in recurrent networks preserve stronger statistical dependencies among hidden representations. Nevertheless, despite the higher baseline integration, the RNN still follows the same overall decreasing trend, indicating that gradient-based optimization continues to drive progressive specialization and redundancy reduction.

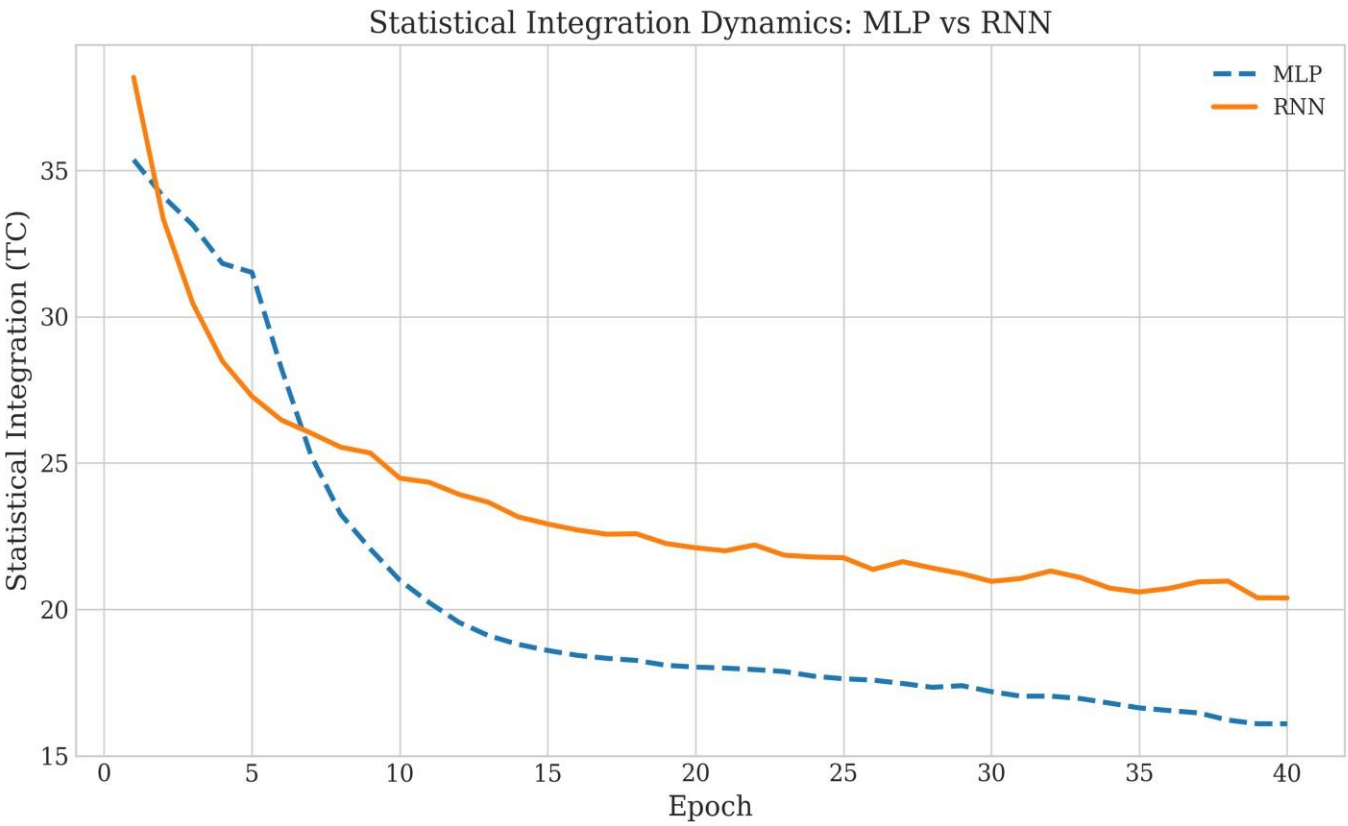

Figure 3. Statistical Integration Dynamics across Architectures (MLP vs RNN)

As shown in Figure 3, the recurrent architecture consistently maintains higher TC values throughout training compared to the feed-forward model, despite both models exhibiting progressive integration decay.

4.4 Temporal Integration Dynamics (RNN)

We conducted exploratory experiments using a simplified proxy for Φ, based on binarized states and an approximate transition probability matrix. Across all experiments, the estimated Φ remained near zero. This result is expected given the limitations of the approximation, including: simplified transition dynamics, lack of causal interventions, and limited temporal depth.

Therefore, we do not interpret these results as evidence about causal integrated information. Instead, we focus on TC as a statistical measure of dependency.

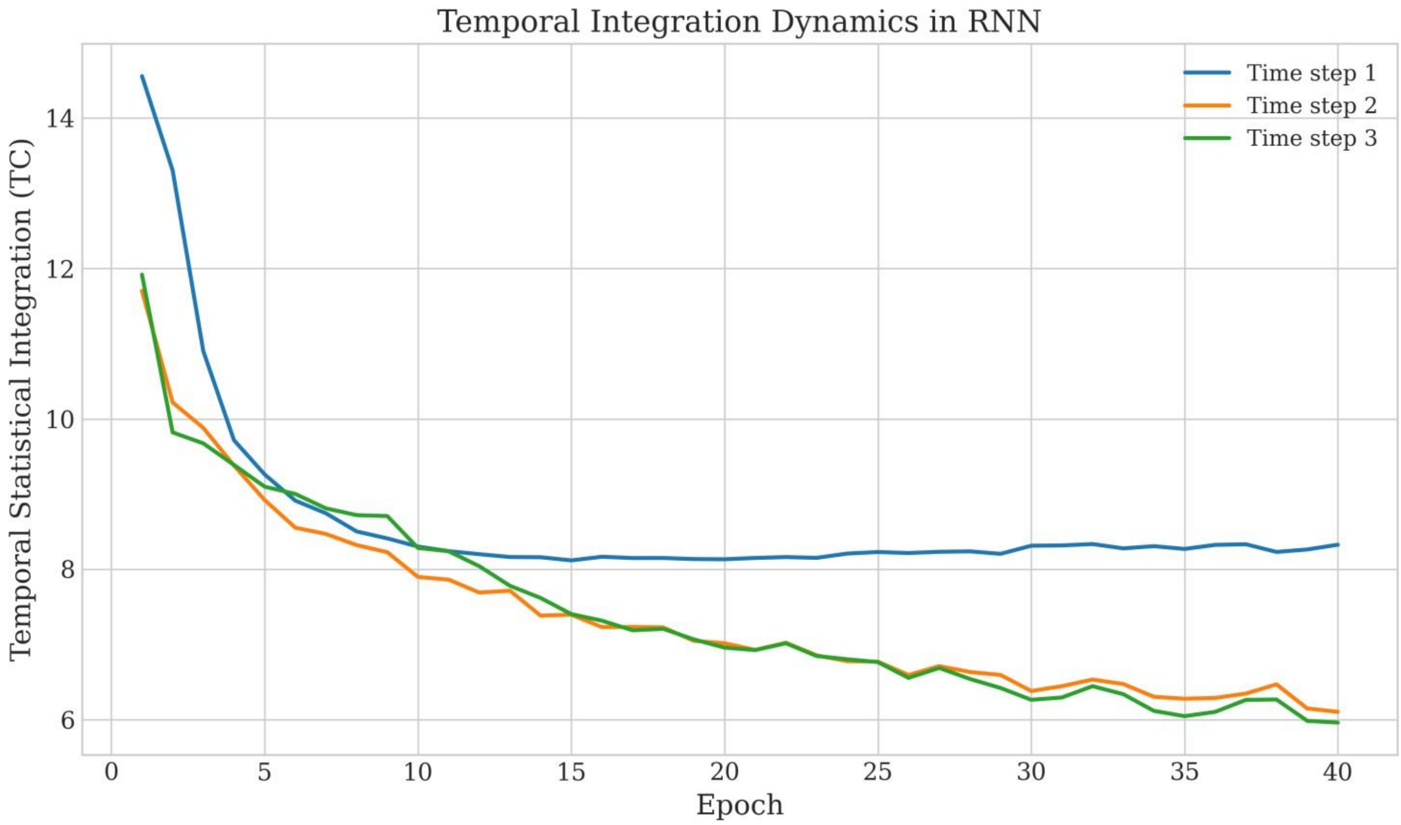

Figure 4. Temporal Integration Dynamics in RNN

Figure 4 shows that later time steps retain slightly higher TC values, indicating that temporal feedback introduces dependencies across time.

4.5 Correlation Analysis

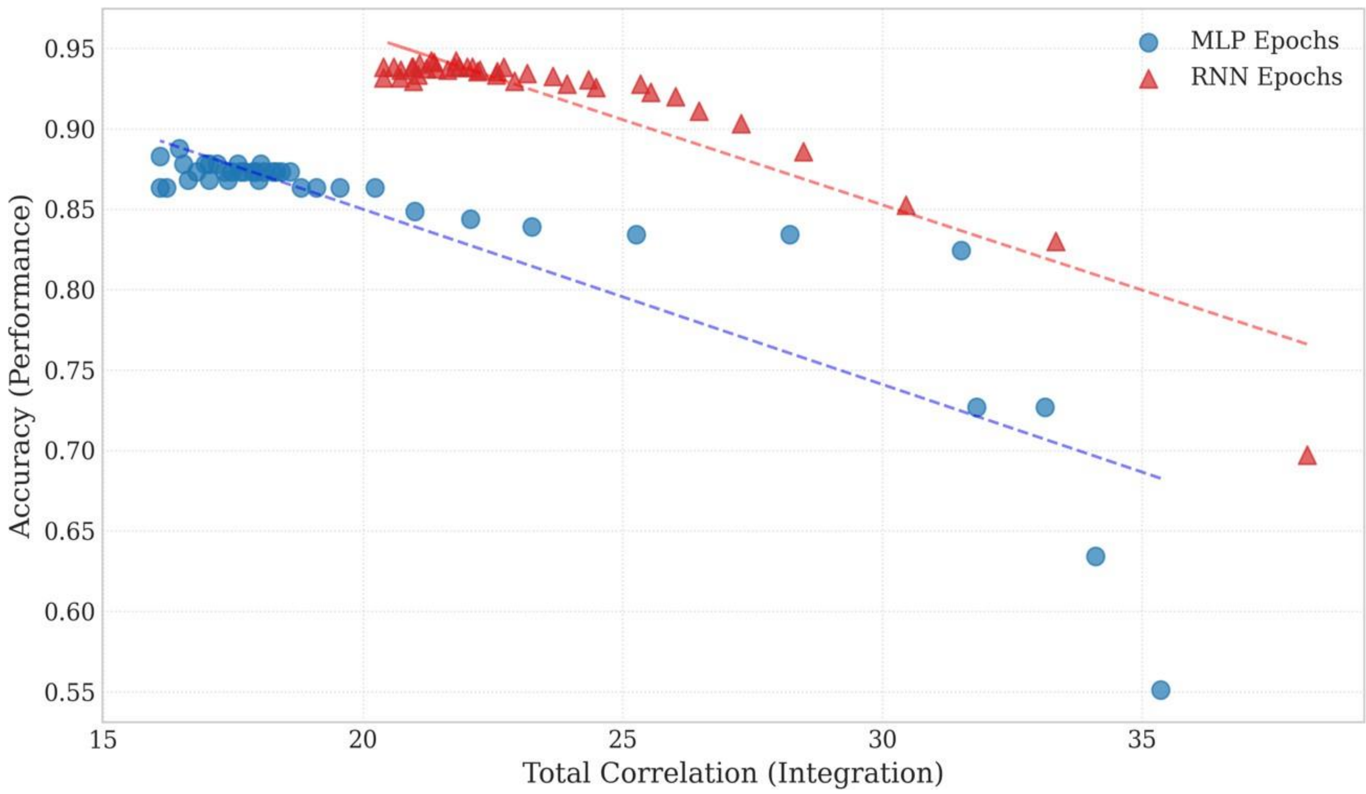

Figure 5. Accuracy vs Total Correlation (TC) showing inverse relationship

Figure 5 illustrates the relationship between classification accuracy and Total Correlation (TC) across training epochs for both architectures. The distributions exhibit a clear negative linear trend, indicating that higher performance is consistently associated with lower statistical integration. The Pearson correlation between accuracy and TC is strongly negative in both models, supporting the hypothesis that gradient-based optimization progressively decorrelates internal representations during learning.

4.6 Summary of Findings

The key empirical observations are: Accuracy increases are consistently associated with reductions in statistical integration (TC). Recurrent architectures maintain higher levels of statistical integration than feed-forward models.

Despite this difference, both architectures exhibit a similar decreasing trend in TC during training. Strong negative correlations between accuracy and TC are observed across epochs. Overall, these results suggest that standard training objectives are associated with progressively decorrelated internal representations, indicating a potential tension between task performance and statistical integration.

Finally, we summarize the overall change in statistical integration from initialization to convergence.

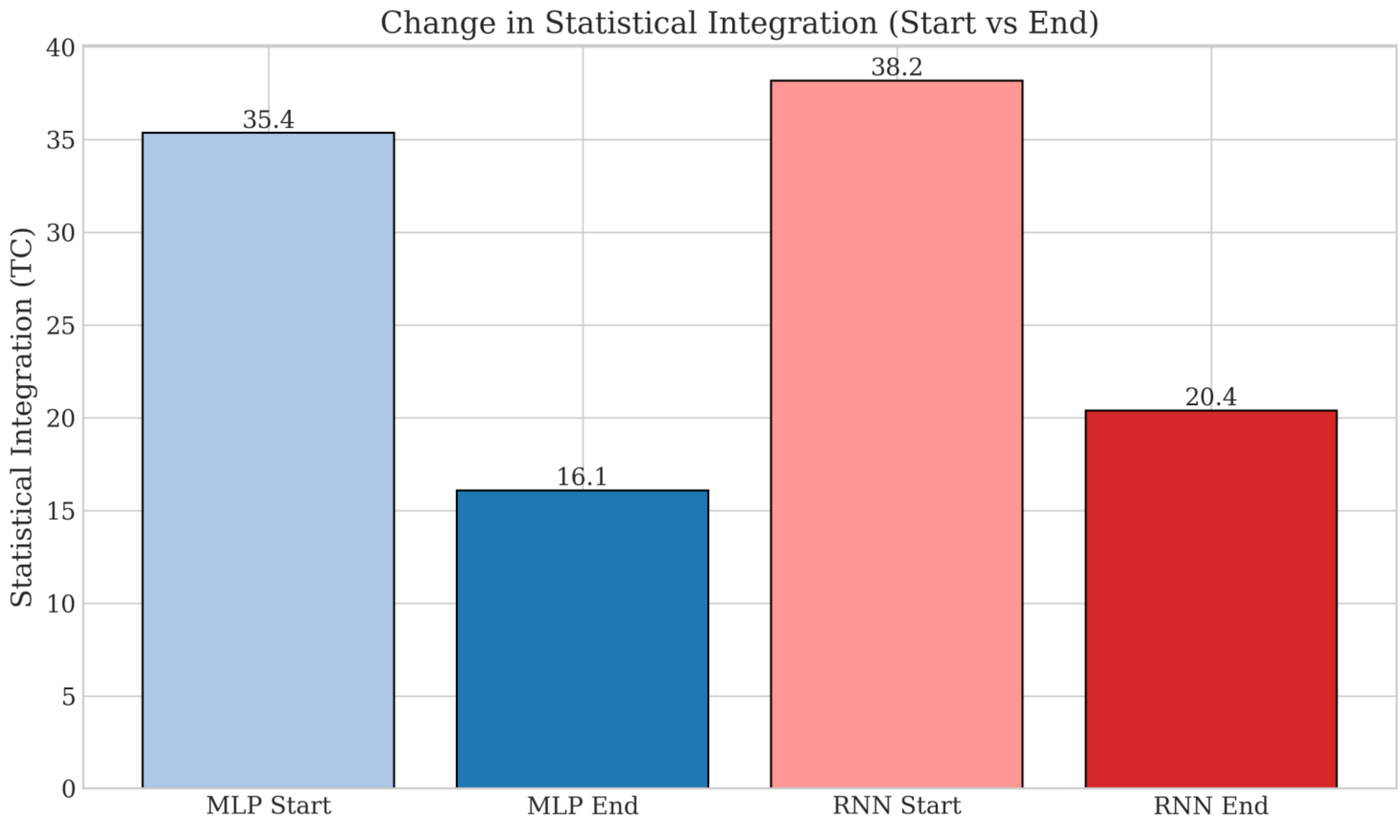

Figure 6. Change in Statistical Integration (Start vs End)

Figure 6 highlights the magnitude of integration reduction in both architectures. While the RNN retains higher final TC values, both models exhibit substantial decreases, reinforcing the general decorrelation trend.

5. DISCUSSION

5.1 Learning as Progressive Decorrelation

The observed inverse relationship between accuracy and Total Correlation (TC) suggests that neural networks learn by progressively reducing statistical dependency among internal representations. At initialization, hidden units exhibit high levels of redundancy, indicating strong statistical coupling.

As optimization proceeds, neurons become increasingly specialized, leading to improved task performance and reduced redundancy. This behavior is consistent with the Information Bottleneck perspective, where effective representations retain only task-relevant information. In this context, our results provide an empirical view of how compression manifests as decorrelation in neural activations [3], [5].

5.2 The Role of Recurrence

The comparison between feed-forward and recurrent architectures highlights the impact of structural design on integration dynamics. Recurrent networks consistently maintain higher levels of statistical integration throughout training.

This is likely due to temporal feedback, which introduces dependencies across time steps and constrains the independence of hidden representations.
However, despite this difference, both architectures exhibit a similar decreasing trend in TC as performance improves. This suggests that while recurrence may support higher baseline integration, it does not eliminate the overall tendency toward decorrelation under standard training objectives.

5.3 Statistical vs Causal Integration

A key limitation of this study is the distinction between statistical dependency (measured by TC) and causal integration (as defined in IIT). TC captures redundancy and dependency among variables, whereas Φ aims to quantify causal irreducibility.

Our observation that TC decreases during training does not directly imply changes in causal integrated information [6], [7]. Therefore, the results should be interpreted as evidence about statistical integration dynamics, not about consciousness or causal structure.

5.4 Limitations

This study has several limitations. First, Gaussian Total Correlation measures statistical dependency rather than causal integrated information in the IIT sense. Second, experiments were conducted on relatively small-scale architectures and simplified benchmark tasks (e.g., XOR).
Third, covariance-based Gaussian approximations may not fully capture non-linear dependency structures in neural representations. Future work should investigate larger architectures and causal measures of integration.

6. CONCLUSION

In this work, we analyzed the relationship between task optimization and statistical integration in neural networks by tracking Gaussian Total Correlation (TC) during training. Our results show that improvements in task performance are consistently associated with reductions in statistical dependency among hidden units.
This pattern is observed across both feed-forward and recurrent architectures. Recurrent networks maintain higher levels of statistical integration throughout training, likely due to temporal feedback mechanisms, but still exhibit a similar decreasing trend. These findings suggest that standard gradient-based optimization is associated with progressively decorrelated internal representations. This highlights a potential tension between task efficiency and statistical integration within current neural architectures.
Future work should investigate whether alternative training objectives or architectural modifications can preserve statistical dependency while maintaining performance.

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